

Supplementary for Connectivity-Aware Saliency Enhances Segmentation of Inconspicuous Liver Vessels by Neural Networks

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Figure S1. The aggregated heatmaps of each evaluation metric across the independent three datasets.

Figure S2. The intra-group violin plots between models with and without the saliency module across all evaluation metrics on 3D-IRCADb.

Figure S3. The intra-group violin plots between models with and without the saliency module across all evaluation metrics on MSD.

Figure S4. The intra-group violin plots between models with and without the saliency module across all evaluation metrics on LiVS.

Figure S5. The intra-group violin plots between models with and without the saliency module across all evaluation metrics on Mixed and multi-level Noised datasets.

Figure S6. The multi-group violin plots to compare different integration protocols across all evaluation metrics under the proposed saliency module on 3D-IRCADb.

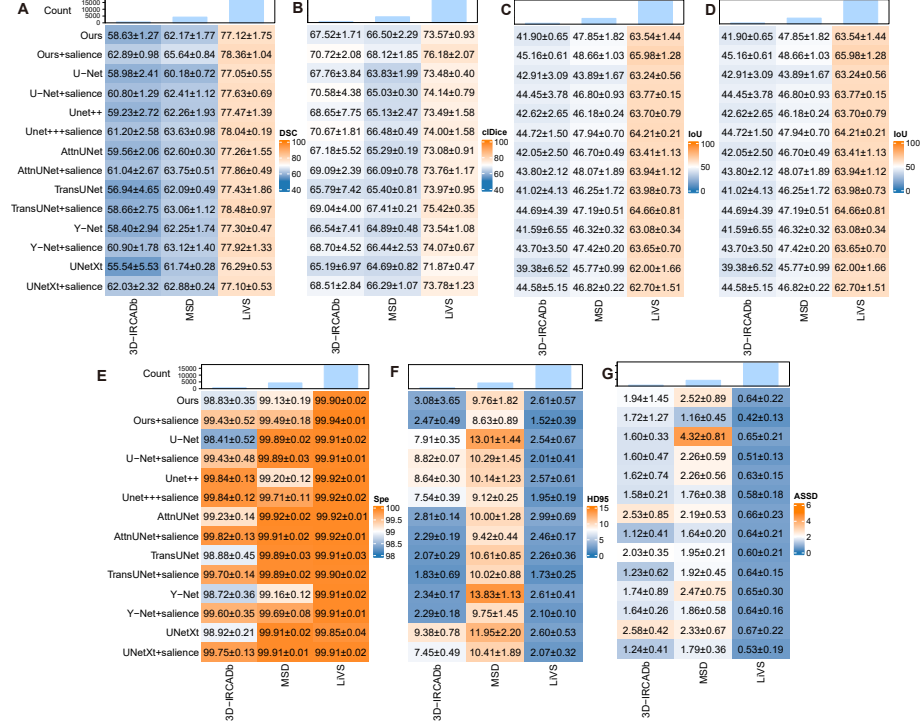


Figure S1. Aggregated heatmaps of each evaluation metric across the independent three datasets, comparing state-of-the-art models with and without the proposed connectivity-aware saliency module. The color depth represents the magnitude of average performance for each setting, and the top histogram indicates the absolute size of the test sets in each dataset. (A) Heatmap of the DSC; (B) Heatmap of the cDice; (C) Heatmap of the IoU; (D) Heatmap of the Sensitivity; (E) Heatmap of the Specificity; (F) Heatmap of the HD95; (G) Heatmap of the ASSD.

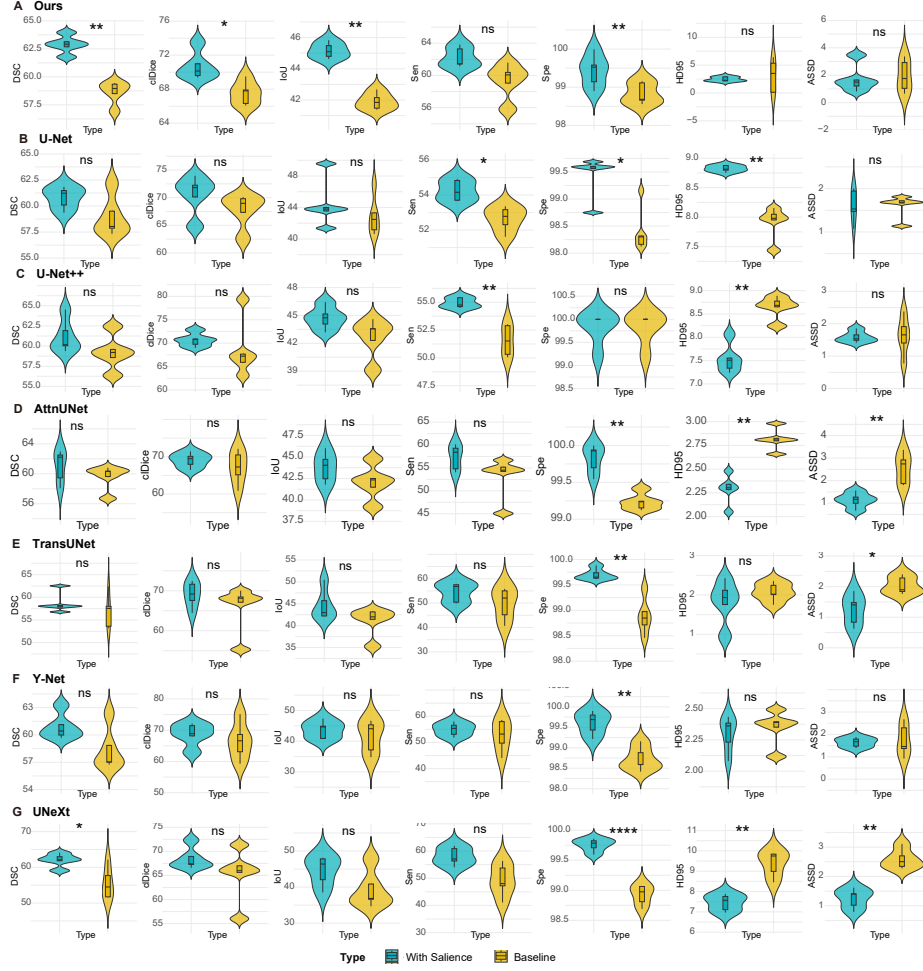


Figure S2. The intra-group violin plots, generated from 5 test samples, compare models with and without the salience module across all evaluation metrics on the 3D-IRCADb dataset. (A-G) show results for the models across Ours, U-Net, U-Net++, AttnUNet, TransUNet, Y-Net, and UNeXt, respectively. Statistical significance: p -value < 0.05(*), p -value < 0.01(**), p -value < 0.0001(****), ns: no statistical significance.

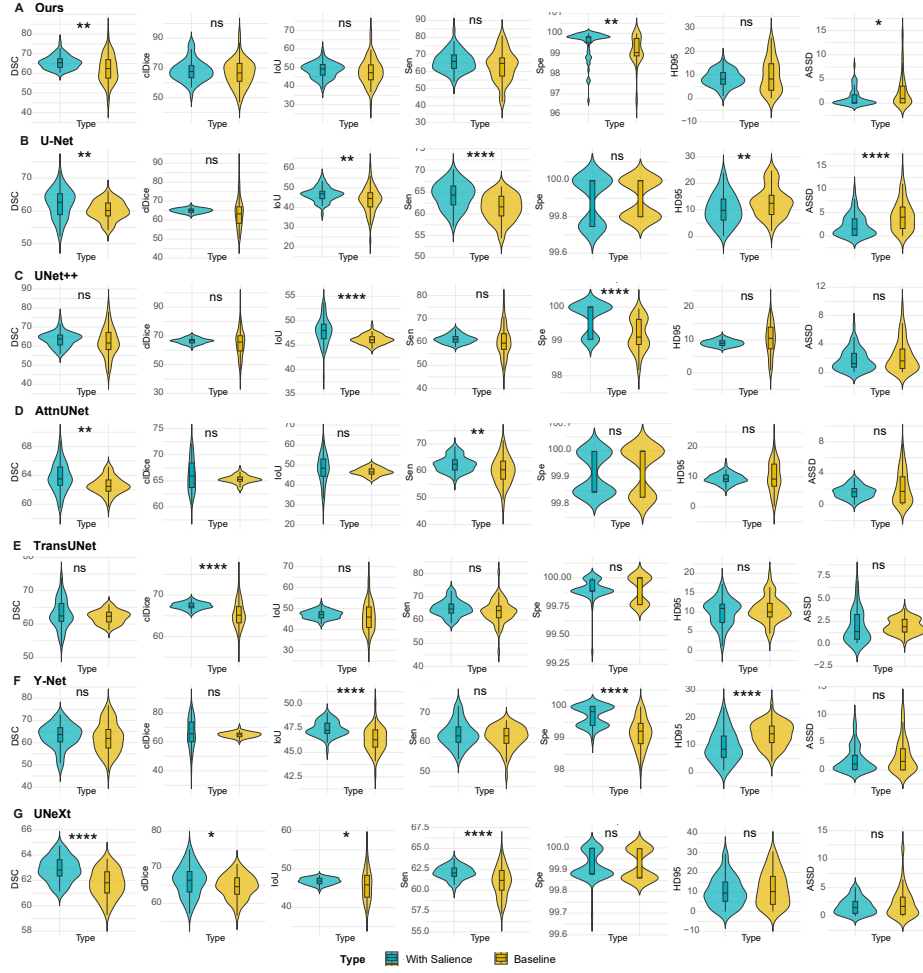


Figure S3. The intra-group violin plots, generated from 61 test samples, compare models with and without the salience module across all evaluation metrics on the MSD. (A-G) show results for the models across Ours, U-Net, U-Net++, AttnUNet, TransUNet, Y-Net, and UNeXt, respectively. Statistical significance: p -value < 0.05(*), p -value < 0.01(**), p -value < 0.0001(****), ns: no statistical significance.

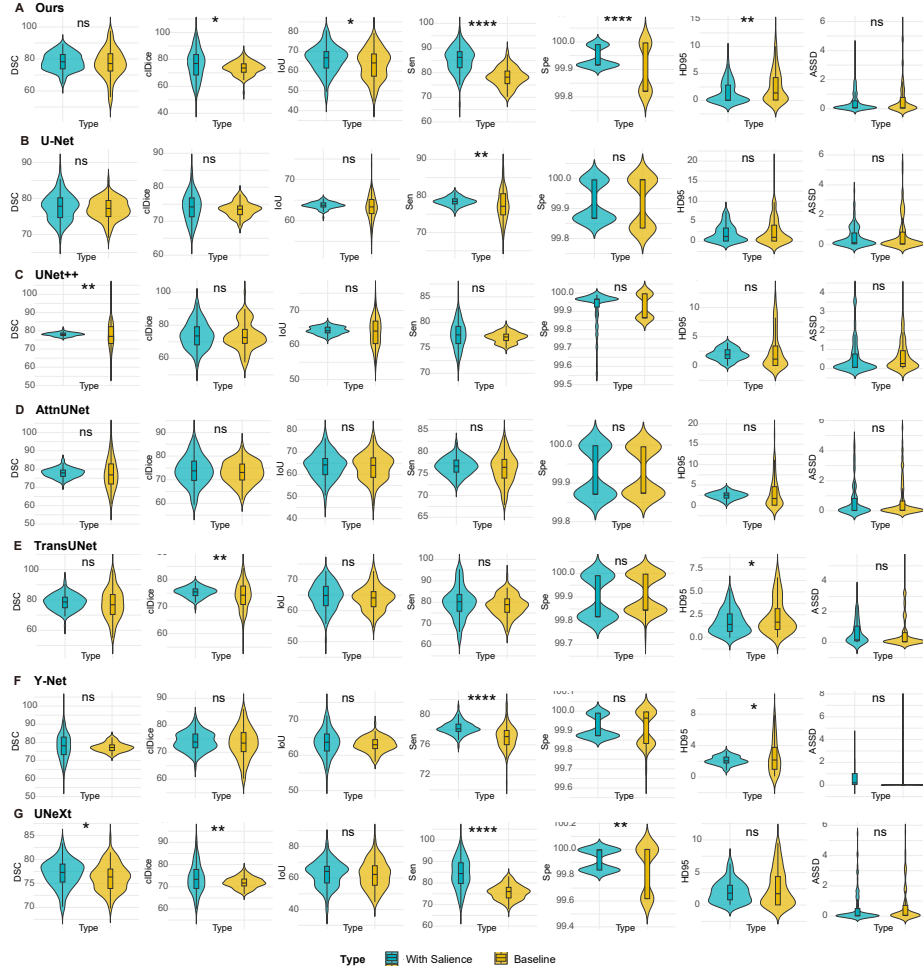


Figure S4. The intra-group violin plots, generated from 106 test samples, compare models with and without the salience module across all evaluation metrics on the LiVS. (A-G) show results for the models across Ours, U-Net, U-Net++, AttnUNet, TransUNet, Y-Net, and UNeXt, respectively. Statistical significance: p -value < 0.05 (*), p -value < 0.01 (**), p -value < 0.0001 (***), ns: no statistical significance.

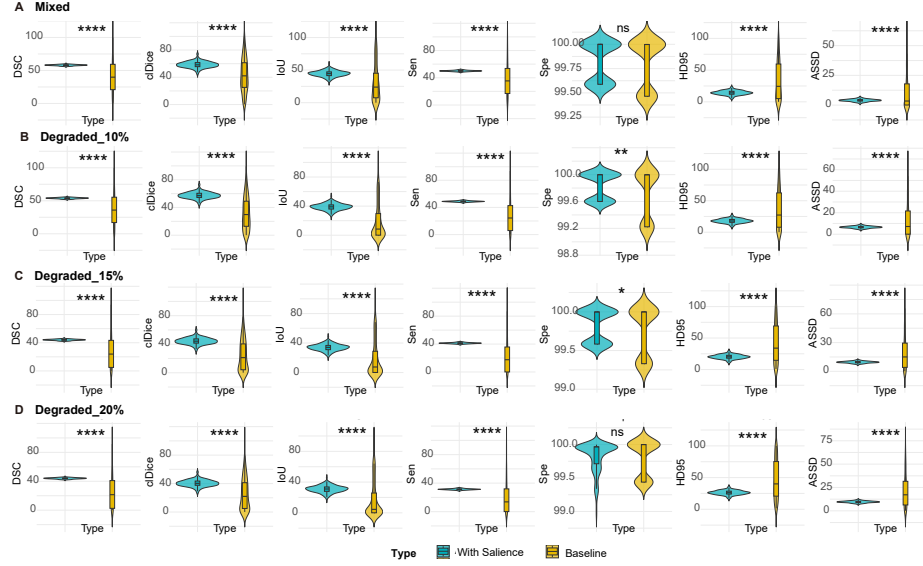


Figure S5. The intra-group violin plots, generated from 172 test samples, compare our baseline with and without the saliency module across all evaluation metrics on Mixed and degraded datasets. (A-D) show results for the models across Mixed, Degraded, Degraded_10%, Degraded_15%, and Degraded_20%, respectively. Statistical significance: p -value < 0.05 (*), p -value < 0.01 (**), p -value < 0.0001 (***), ns: no statistical significance.

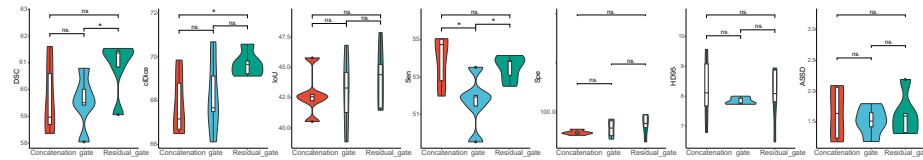


Figure S6. The multi-group violin plots, generated from 5 test samples, compare different integration protocols across all evaluation metrics under the proposed saliency module on the 3D-IRCADb dataset. The horizontal axis represents the integration protocols (Concatenation, Gate, and Residual Gate), and the vertical axis indicates the corresponding metric values. Statistical significance: p -value < 0.05 (*), ns: no statistical significance.